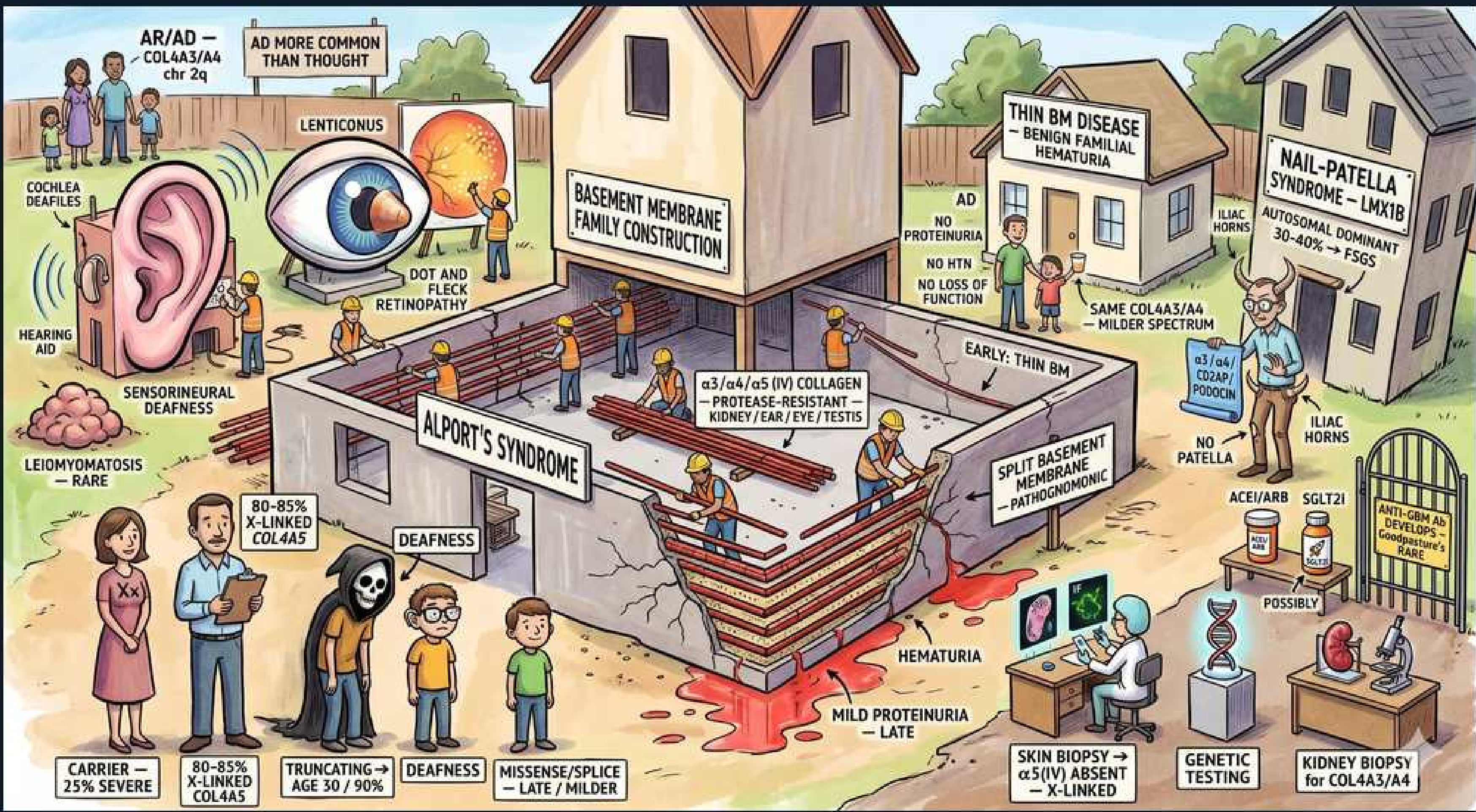
Glomerular (Nephrotic) — Basement Membrane Syndromes: Alport / Thin BM / Nail-Patella



1. Alport's syndrome banner

WHAT YOU SEE

Central building beam labeled ALPORT'S SYNDROME under 'Basement Membrane Family Construction'

WHAT IT ENCODES

Alport syndrome (hereditary nephritis) is a STRUCTURAL defect of type IV collagen in the glomerular basement membrane, not an immune disease. The classic triad is hereditary nephritis (progressive hematuria → proteinuria → ESRD) + bilateral high-frequency SENSORINEURAL deafness + OCULAR changes (anterior lenticonus, dot-and-fleck retinopathy). Mnemonic: 'can't SEE, can't HEAR, can't PEE.' It presents as a NEPHRITIC picture (hematuria) early despite living on the nephrotic-syndrome page.

BOARD TRAP

WRONG: classifying Alport as an immune-complex glomerulonephritis or a primarily nephrotic disease. The discriminator is mechanism — Alport is a hereditary STRUCTURAL collagen IV defect (negative immunofluorescence, no immune complexes), and it presents NEPHRITIC with hematuria, not with early heavy nephrotic-range proteinuria.

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2. COL4A3/4/5 collagen

WHAT YOU SEE

Beam reading α3/α4/α5(IV) collagen — protease-resistant — kidney/ear/eye/testis

WHAT IT ENCODES

The defect is in type IV collagen genes COL4A3, COL4A4, or COL4A5, encoding the α3/α4/α5(IV) heterotrimer that forms the mature, protease-resistant collagen network of the GBM, cochlea, lens, and testis. Loss of this network leaves a weaker α1/α2(IV) scaffold that splits and lamellates over time — explaining the multi-organ (kidney/ear/eye) involvement and the progressive nature.

BOARD TRAP

WRONG: pinning Alport on type III collagen or on anti-GBM antibody. The discriminator is gene vs antigen — Alport is a GENE defect in type IV collagen (COL4A3/4/5), distinct from Goodpasture/anti-GBM disease, which is an autoantibody against the α3(IV) NC1 antigen; the shared α3(IV) target is why post-transplant Alport patients can rarely develop anti-GBM disease against the now-foreign normal collagen.

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3. X-linked COL4A5

WHAT YOU SEE

Family/skeleton panel '80-85% X-linked COL4A5'

WHAT IT ENCODES

The most common form (~80–85%) is X-LINKED via COL4A5 mutation → MALES are severely affected (progress to ESRD, often by their 20s–40s), while heterozygous FEMALES are usually milder carriers with hematuria and variable later risk. Autosomal recessive (COL4A3/COL4A4) forms are severe and affect both sexes; autosomal dominant forms are milder. X-linked inheritance pattern (no male-to-male transmission, affected sons of carrier mothers) is the classic pedigree clue.

BOARD TRAP

WRONG: expecting autosomal dominant inheritance or male-to-male transmission as the typical pattern. The discriminator is X-LINKED dominance — the majority are X-linked COL4A5 with characteristic absence of father-to-son transmission and severe disease in males; AR (COL4A3/4) forms exist and are severe but are the minority.

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4. Sensorineural deafness

WHAT YOU SEE

Ear with hearing aid, 'COCHLEA deafness / sensorineural deafness'

WHAT IT ENCODES

Bilateral high-frequency SENSORINEURAL hearing loss is a hallmark, caused by the same defective α3/α4/α5(IV) collagen in the cochlear basement membranes. It is progressive, typically becomes apparent in late childhood/adolescence, and correlates with renal severity. Pairing renal hematuria with sensorineural deafness in a young male is the classic Alport vignette.

BOARD TRAP

WRONG: labeling the hearing loss conductive. The discriminator is the type — Alport causes SENSORINEURAL (cochlear/collagen-based) hearing loss, not conductive; a conductive loss points away from Alport toward middle-ear pathology.

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5. Lenticonus / eye

WHAT YOU SEE

Eye panel with LENTICONUS and dot-and-fleck retinopathy

WHAT IT ENCODES

Ocular findings stem from defective lens-capsule and retinal collagen: ANTERIOR LENTICONUS (conical protrusion of the lens, considered pathognomonic for Alport) and dot-and-fleck retinopathy (yellowish perimacular flecks). Anterior lenticonus causes progressive myopia/lens distortion and, when present, is essentially diagnostic of Alport.

BOARD TRAP

WRONG: calling the ocular finding a cataract. The discriminator is the lesion — ANTERIOR LENTICONUS (lens-capsule cone) is specific to Alport, whereas a simple cataract (lens opacity) is nonspecific and does not point to a collagen IV disorder.

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6. Hematuria

WHAT YOU SEE

Red blood pooling at base labeled HEMATURIA

WHAT IT ENCODES

Persistent microscopic HEMATURIA is the earliest and essentially universal renal sign, often present from childhood; episodes of GROSS hematuria may follow upper respiratory infections in children. Glomerular bleeding gives dysmorphic RBCs and RBC casts. Hematuria, not proteinuria, dominates the early clinical picture — this is the nephritic face of Alport.

BOARD TRAP

WRONG: expecting isolated proteinuria as the presenting feature. The discriminator is that HEMATURIA (microscopic, persistent, often from childhood) dominates early; significant proteinuria is a LATER development, so an early picture of isolated heavy proteinuria argues against Alport.

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7. Mild proteinuria late

WHAT YOU SEE

Lower banner 'MILD PROTEINURIA — LATE'

WHAT IT ENCODES

Proteinuria appears LATER as the splitting/lamellated GBM progressively fails, and it heralds the slide toward progressive renal failure and ESRD (especially in affected males and AR forms). It can reach significant levels over time but is not the early hallmark. Early treatment with ACE inhibitors/ARBs to reduce proteinuria slows progression to ESRD — the key disease-modifying intervention.

BOARD TRAP

WRONG: anticipating early heavy nephrotic-range proteinuria. The discriminator is the time course — proteinuria in Alport is a LATE feature signaling disease progression, not the initial presentation; early on the urine shows hematuria with little or no protein, and starting RAAS blockade early is what slows ESRD.

8. Thin BM disease

WHAT YOU SEE

Right panel THIN BM DISEASE — benign familial hematuria

WHAT IT ENCODES

Thin basement membrane nephropathy (benign familial hematuria) presents as ISOLATED persistent microscopic hematuria with NORMAL renal function, normal blood pressure, and no significant proteinuria — a benign course. It is the most common cause of persistent isolated glomerular hematuria. On EM the GBM is diffusely and uniformly THIN (rather than lamellated/split as in Alport).

BOARD TRAP

WRONG: equating thin BM disease with Alport because both have hematuria and COL4 involvement. The discriminator is PROGNOSIS and EM — thin BM disease is BENIGN (no progression to ESRD, uniformly thin GBM), whereas Alport progresses to ESRD with a lamellated 'basket-weave' GBM and extra-renal deafness/eye findings.

9. Thin BM: AD, no progression

WHAT YOU SEE

Stick-figure panel 'AD / no proteinuria / no HTN / no loss of function'

WHAT IT ENCODES

Thin BM nephropathy is typically AUTOSOMAL DOMINANT, frequently due to a HETEROZYGOUS COL4A3/COL4A4 mutation. The hallmark is benignity: NO hypertension, NO significant proteinuria, and PRESERVED GFR over the lifetime. Management is REASSURANCE plus periodic monitoring of BP, urine protein, and renal function — not immunosuppression.

BOARD TRAP

WRONG: treating thin BM disease with immunosuppression. The discriminator is that it is a benign structural variant requiring only REASSURANCE and monitoring — immunosuppression is inappropriate (there is no immune-mediated injury to suppress) and exposes the patient to needless harm.

10. Same COL4A3/4 spectrum

WHAT YOU SEE

Note 'same COL4A3/A4 — milder spectrum / early: thin BM'

WHAT IT ENCODES

Thin BM nephropathy and autosomal Alport sit on the SAME COL4A3/COL4A4 disease spectrum — thin BM (often heterozygous) is the mild end, and homozygous/compound mutations produce autosomal recessive Alport at the severe end. This is why a heterozygous CARRIER of AR Alport may manifest only as thin-BM-type isolated hematuria. The continuum reframes these as one genetic family, not unrelated diseases.

BOARD TRAP

WRONG: treating thin BM disease and autosomal Alport as genetically separate, so a heterozygous carrier's hematuria is dismissed as wholly benign. The discriminator is the SHARED COL4A3/A4 spectrum — carriers of AR Alport can present as thin-BM hematuria, so family history and genetic testing matter because some 'benign' hematuria carriers harbor risk within an Alport family.

11. Nail-Patella syndrome

WHAT YOU SEE

Top-right NAIL-PATELLA SYNDROME — LMX1B, autosomal dominant

WHAT IT ENCODES

Nail-patella syndrome is an AUTOSOMAL DOMINANT disorder from LMX1B mutation (a transcription factor controlling dorsoventral limb patterning and podocyte/GBM development). Features: dysplastic/absent NAILS, absent or hypoplastic PATELLAE, ILIAC HORNS, elbow dysplasia, and renal involvement (proteinuria, sometimes nephrotic, with an FSGS-like lesion). On EM the GBM shows characteristic 'moth-eaten' lucencies with collagen fibrils.

BOARD TRAP

WRONG: grouping nail-patella with the type IV collagen diseases. The discriminator is the gene — nail-patella is caused by LMX1B (a transcription factor), NOT a COL4 collagen gene, so it sits alongside Alport/thin-BM in the basement-membrane 'family' visually but has a distinct genetic mechanism and skeletal/nail phenotype.

12. Iliac horns / no patella

WHAT YOU SEE

Skeleton with 'ILIAC HORNS' and 'NO PATELLA'

WHAT IT ENCODES

The skeletal signature of nail-patella syndrome is bilateral posterior ILIAC HORNS (bony spurs off the iliac bones, pathognomonic on pelvic X-ray) plus absent or hypoplastic PATELLAE (causing recurrent knee instability/dislocation). These radiographic clues, combined with nail dysplasia, clinch the diagnosis even before renal involvement appears.

BOARD TRAP

WRONG: overlooking iliac horns as the diagnostic clue and chasing the renal lesion alone. The discriminator is the skeletal finding — ILIAC HORNS are essentially pathognomonic for nail-patella syndrome (seen in no other condition), so a pelvic film showing them, plus absent patellae and dysplastic nails, makes the diagnosis.

13. Skin biopsy α5(IV) absent

WHAT YOU SEE

Microscope panel 'skin biopsy → α5(IV) absent → X-linked'

WHAT IT ENCODES

Because α5(IV) collagen is normally expressed in EPIDERMAL basement membrane too, a SKIN biopsy showing ABSENT α5(IV) staining confirms X-LINKED Alport (COL4A5) — a quick, less invasive diagnostic shortcut than renal biopsy. Genetic testing (COL4A5/COL4A3/COL4A4 sequencing) is increasingly the confirmatory and first-line diagnostic tool.

BOARD TRAP

WRONG: requiring a renal biopsy when a skin biopsy or genetic test would confirm Alport. The discriminator is that absent EPIDERMAL α5(IV) staining on skin biopsy diagnoses X-linked Alport non-invasively; a NORMAL skin α5(IV) does not exclude autosomal (COL4A3/4) Alport, so genetic testing is the definitive arbiter across all forms.

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14. Kidney biopsy / EM basket-weave

WHAT YOU SEE

Right 'kidney biopsy for COL4A3/A4' / pathognomonic note

WHAT IT ENCODES

Renal biopsy EM is diagnostic: Alport shows the pathognomonic LAMELLATED 'BASKET-WEAVE' splitting of the GBM with alternating thick and thin segments, in contrast to the diffusely and UNIFORMLY THIN GBM of benign thin BM disease. Immunofluorescence is negative for immune deposits (this is structural, not immune-complex disease).

BOARD TRAP

WRONG: confusing the EM patterns of Alport and thin BM disease. The discriminator is GBM architecture on EM — Alport = lamellated/split 'BASKET-WEAVE' GBM (progresses to ESRD), whereas thin BM disease = diffusely UNIFORM thinning (benign); both can involve COL4 but the EM appearance and prognosis diverge sharply.
